WORKED SOLUTION

STEP II 2006, Question 5

Pure • Floor Function Areas

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s2-q5

The key idea

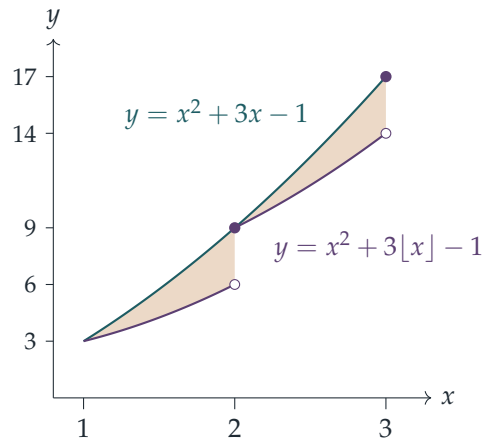
Between two consecutive integers $k \leq x < k+1$ the floor function is constant, equal to k. So on each such unit interval the jagged curve is an ordinary, smooth curve, and the area splits into a sum of strips, one per unit interval. Work out the area of the strip over $[k, k+1]$ as a function of k, then add up the strips for $k = 1, 2, \ldots, n-1$.

Solution

Throughout, write $f(x) = x^2 + 3x - 1$. On $1 \leq x \leq 3$ it rises from $f(1) = 3$ through $f(2) = 9$ to $f(3) = 17$, and it is increasing for all $x \geq 1$ since $f'(x) = 2x + 3 > 0$ there.

Part (i)

The second curve is $y = x^2 + 3\lfloor x \rfloor - 1$. On $[1, 2)$ we have $\lfloor x \rfloor = 1$, so it is $y = x^2 + 2$, rising from 3 towards 6; on $[2, 3)$ we have $\lfloor x \rfloor = 2$, so it is $y = x^2 + 5$, rising from 9 towards 14; and at $x = 3$ it equals 17. Each piece is the parabola $y = f(x)$ shifted down, and the two curves meet at every integer.



The vertical gap between the curves is

$$(x^2 + 3x - 1) - (x^2 + 3\lfloor x \rfloor - 1) = 3(x - \lfloor x \rfloor) \geq 0,$$

so the first curve is never below the second. On the interval $[k, k+1]$, with k an integer, the gap is $3(x-k)$ except at the single point $x = k+1$, which does not affect an integral. So the strip over $[k, k+1]$ has area

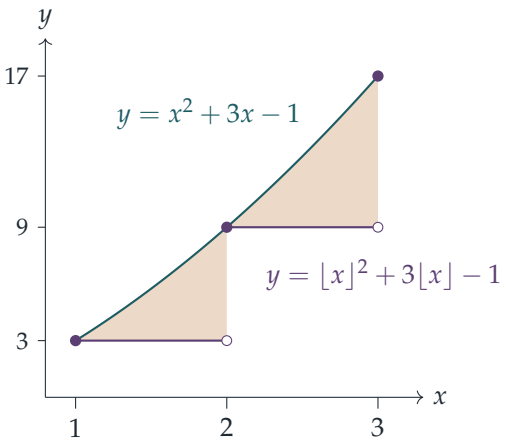
$$\int_k^{k+1} 3(x-k)\,\mathrm{d}x = \left[\tfrac{3}{2}(x-k)^2\right]_k^{k+1} = \tfrac{3}{2},$$

the same for every k. The interval $1 \leq x \leq n$ is made up of the $n-1$ unit intervals $[1,2], [2,3], \ldots, [n-1, n]$, so the area between the curves is

$$\tfrac{3}{2}(n-1).$$

Part (ii)

Now the second curve is $y = f(\lfloor x \rfloor)$, a step function: $y = f(1) = 3$ on $[1,2)$, $y = f(2) = 9$ on $[2,3)$, and $y = f(3) = 17$ at $x = 3$. Each step starts on the parabola at an integer and runs horizontally to the right.



Ask yourself Which curve is on top, and does it stay on top?

Since f is increasing for $x \geq 1$ and $\lfloor x \rfloor \leq x$, we have $f(\lfloor x \rfloor) \leq f(x)$ throughout $1 \leq x \leq n$. So the parabola is always on top and the area is simply the integral of $f(x) - f(\lfloor x \rfloor)$, with no sign changes to worry about.

On $[k, k+1]$, where $k \geq 1$ is an integer, the step sits at height $f(k)$ (again ignoring the single point $x = k+1$). Substituting $x = k + t$ with $0 \leq t \leq 1$,

$$\begin{aligned} f(k+t) - f(k) &= (k+t)^2 + 3(k+t) - 1 - (k^2 + 3k - 1) \\ &= 2kt + t^2 + 3t, \end{aligned}$$

so the strip over $[k, k+1]$ has area

$$\int_0^1 (2kt + t^2 + 3t)\,\mathrm{d}t = k + \tfrac{1}{3} + \tfrac{3}{2} = k + \tfrac{11}{6}.$$

Unlike part (i), the strips now grow with k, because the parabola gets steeper. Adding the strips for $k = 1, 2, \ldots, n-1$ and using $\sum_{k=1}^{n-1} k = \frac{1}{2}n(n-1)$,

$$\begin{aligned} \text{Area} &= \sum_{k=1}^{n-1} \left(k + \tfrac{11}{6}\right) = \tfrac{1}{2}n(n-1) + \tfrac{11}{6}(n-1) \\ &= \tfrac{1}{6}(n-1)(3n+11), \end{aligned}$$

which is the required result.

Checks

- With $n = 1$ there is no region at all, and both answers give 0.
- With $n = 2$, part (ii) gives $\frac{1}{6} \cdot 1 \cdot 17 = \frac{17}{6}$. Directly, $\int_1^2 (x^2 + 3x - 4)\,\mathrm{d}x = \frac{7}{3} + \frac{9}{2} - 4 = \frac{17}{6}$.
- A midpoint-rule integration of the gaps for $n = 2, \ldots, 7$ matched $\frac{3}{2}(n-1)$ and $\frac{1}{6}(n-1)(3n+11)$ to four decimal places; for example, $n = 4$ gives 11.5 in part (ii).

Common slips

- Counting n strips instead of $n-1$: the interval $[1, n]$ contains only $n-1$ unit intervals.
- In part (ii), using $\int_0^1 2kt\,\mathrm{d}t = 2k$ instead of k.
- Integrating $\lfloor x \rfloor$ as if it were x (or ignoring it) over a whole interval, rather than splitting at the integers first.

Beyond STEP

Part (ii) compares the area under a curve with the area of a "staircase" of rectangles under it, which is exactly a lower Riemann sum with step width 1. In a first analysis course, the integral is *defined* by squeezing such lower and upper sums together as the steps get narrower, and functions with jumps, like $\lfloor x \rfloor$, are still integrable because their jumps are isolated points. Abbott's *Understanding Analysis* gives a very readable account of the Riemann integral.

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